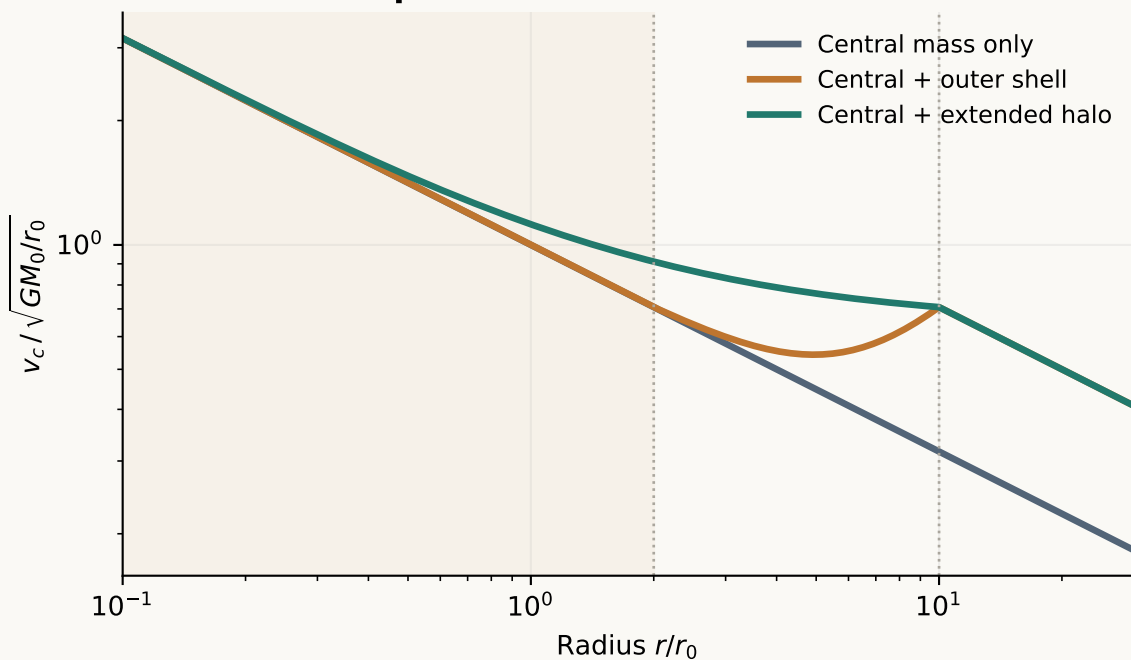
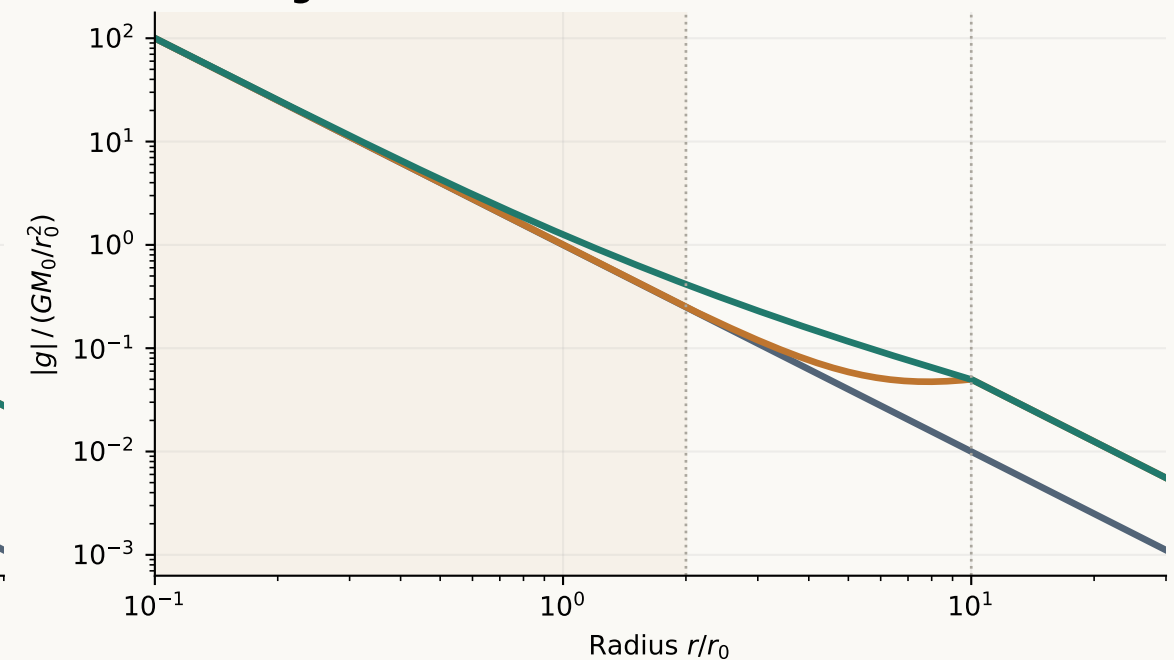


An outer shell and an extended halo are different gravitational models

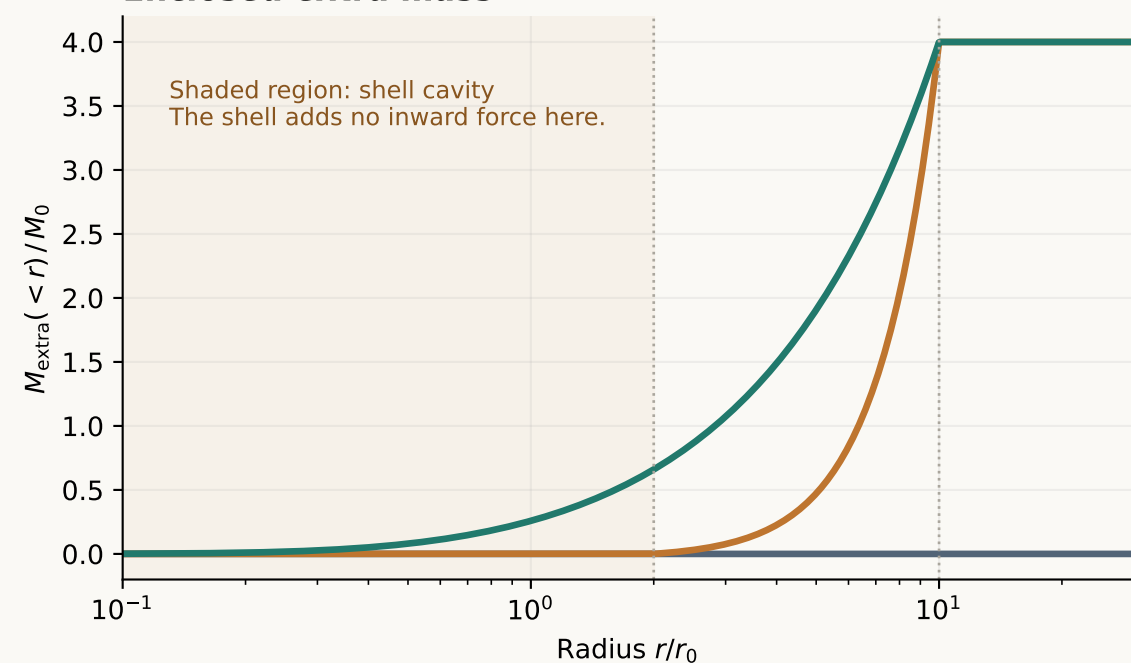
Circular-orbit speed



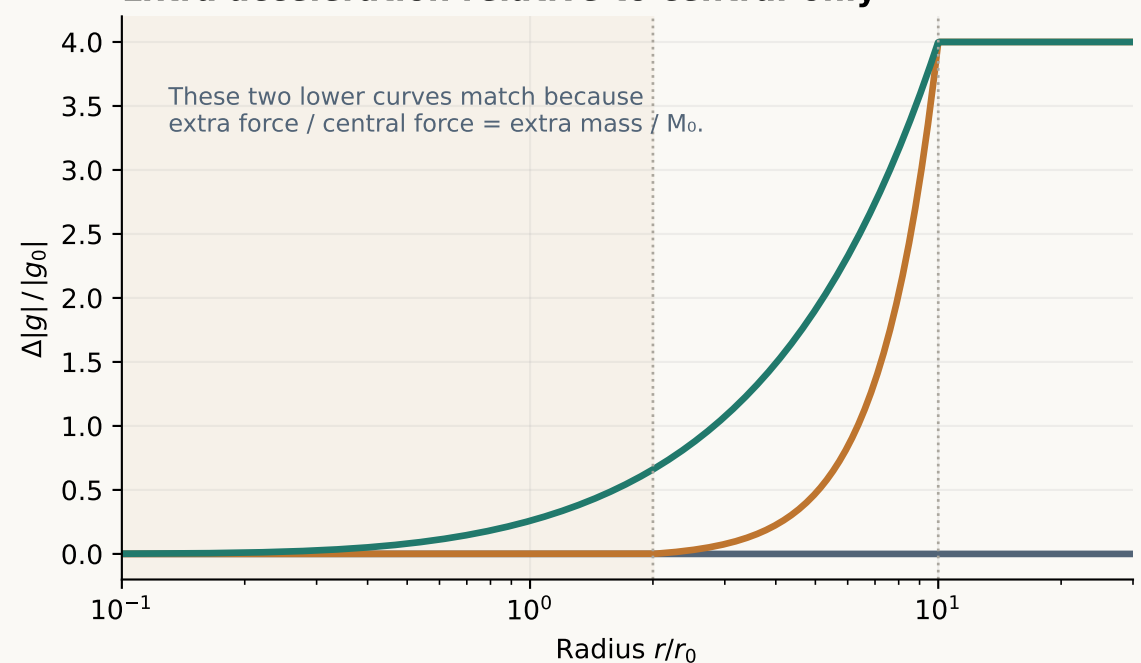
Inward gravitational acceleration



Enclosed extra mass



Extra acceleration relative to central-only



Toy comparison: same extra mass ($4 M_0$), same outer edge ($10 r_0$); uniform shell starts at $2 r_0$.
Cored halo: $\rho \propto 1 / [1 + (r / 0.3r_0)^2]$. All three are Keplerian beyond the outer edge.